

Chapter 21

Lecture 21: Complete regularity. Urysohn's metrization theorem.

Theorem 21.0.1 (Urysohn's Lemma(Recalling)). *Let X be a normal space. If A and B are disjoint closed subsets of X , then there exists a continuous function $f : X \rightarrow [0, 1]$ such that $f(a) = 0$ for all $a \in A$ and $f(b) = 1$ for all $b \in B$.*

Definition 21.0.1. *Suppose X satisfies (T1). If a point and a closed set can be separated by a continuous function as in Urysohn's lemma, then X is called completely regular (CR). (Sometimes it is called $T_{3\frac{1}{2}}$ space.)*

Property 21.0.1. *Every normal space is completely regular.*

Proof By Urysohn's lemma. ■

Remark

Normal $\Rightarrow$ CR $\Rightarrow$ Regular. That's why we call it $T_{3\frac{1}{2}}$ space.

Property 21.0.2. *If X is completely regular, then it is regular.*

Proof

Take $x_0 \in X$ and a closed set A such that $x_0 \notin A$. Since X is CR, there exists a continuous function $f : X \rightarrow [0, 1]$ such that $f(x_0) = 0$ and $f(a) = 1$ for all $a \in A$. Let $U = f^{-1}([0, \frac{1}{2}))$ and $V = f^{-1}((\frac{1}{2}, 1])$. Then U and V are open sets because f is continuous and $[0, \frac{1}{2})$, $(\frac{1}{2}, 1]$ are open in $[0, 1]$. So $x_0 \in U$, $A \subseteq V$ and $U \cap V = \emptyset$. Thus X is regular. ■

Theorem 21.0.2. 1. *Subspace of CR space is CR.*

2. *If X_α is CR, then $\prod X_\alpha$ with product topology is CR.*

Proof

Let X be CR, and $Y \subseteq X$ with subspace topology. Take $y_0 \in Y$ and a closed set $A \subseteq Y$ such that $y_0 \notin A$. Let $\bar{A}$ be the closure of A in X so $A = \bar{A} \cap Y$. Then $\bar{A}$ is closed in X and $y_0 \notin \bar{A}$. Since X is CR, there exists a continuous function $f : X \rightarrow [0, 1]$ such that $f(y_0) = 0$ and $f(a) = 1$ for all $a \in \bar{A}$. Restrict f to Y , we get a continuous function $f|_Y : Y \rightarrow [0, 1]$ such that $f|_Y(y_0) = 0$ and $f|_Y(a) = 1$ for all $a \in A$. Thus Y is CR.

Take $A \subseteq \prod X_\alpha$ closed and $x = (x_\alpha) \notin A$. Then there exists a basic open set $U = \prod U_\alpha$ such that $x \in U$ and $U \cap A = \emptyset$, where U_α is open in X_α and $U_\alpha = X_\alpha$ for all but finitely many α . Let those finitely many α be $\alpha_1, \alpha_2, \dots, \alpha_n$. Since X_{α_i} is CR, there exists a continuous function $f_{\alpha_i} : X_{\alpha_i} \rightarrow [0, 1]$ such that $f_{\alpha_i}(x_{\alpha_i}) = 1$ and $f_{\alpha_i}(a) = 0$ for all $a \in X_{\alpha_i} \setminus U_{\alpha_i}$. So we can define a continuous function $f : \prod X_\alpha \rightarrow [0, 1]$ by

$$f(x) = \prod_{i=1}^n f_{\alpha_i}(\pi_{\alpha_i}(x)).$$

Then $f(x) = 1$ and for all $a \in A$, $f(a) = 0$. Thus $\prod X_\alpha$ is CR. ■

Example 21.0.1. $\mathbb{R}_l$ is normal then it is CR. So $\mathbb{R}_l^2$ is CR. But it is not normal as we have shown.

Fact(no proof) There exist regular spaces that are not CR.

Remark Proof of Urysohn's lemma for metric spaces(exercise):

Let X be metric space with metric $d : X \times X \rightarrow \mathbb{R}$. Given two disjoint closed sets $A, B \subseteq X$, define $d_B(x) = d(x, B)$ which vanishes exactly on B . Then define $d_A(x) = d(x, A)$ which vanishes exactly on A . Now define

$$f(x) = \frac{d_A(x)}{d_A(x) + d_B(x)}.$$

Then $f(A) = \{0\}$ and $f(B) = \{1\}$ and f is continuous because we've proved that distance from a point to a closed set is continuous.

Theorem 21.0.3 (Urysohn Metrization Theorem). *Every regular space with countable basis is metrizable.*

Proof

You can see the proof in P215 of Munkres' book, P232 of the pdf.

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